

MIE375 — Financial Engineering

Course Notes

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Lecture 1: Introduction/Syllabus

September 14, 2026

Principal and Interest

Def. Principal: Amount invested

Interest: "Rent" paid on investment

In general, for

- A : Principal
- r : interest rate per period
- V : Value received.

After 1 period, $V = A(1 + r)$ is received

Interest over multiple periods depend on the compounding rules:

1. **Simple**: Money grows linearly, receiving $V = A(1 + nr)$ over n periods.
2. **Compound**: Money grows geometrically, receiving $V = A(1 + r)^n$ over n periods.

The Rule of 72: Money invested at $r\%$ per year doubles in approximately $72/r$ years.

Def. Nominal Interest Rate: annual quote, compounding interest is in fractions corresponding to the compounding period.

For m periods per year for k periods, $V = (1 + \frac{r}{m})^k$

Def. Continuous Compound Interest: when the compounding period becomes arbitrarily small, the value approaches e^{rt} .

Relationship between Nominal and Effective Rate:

$$1 + r_{eff} = (1 + \frac{r_{nom}}{m})^m$$

$1 + \text{Simple One-year Rate} = \text{the Explicit Compound}$

Lecture 2: Present Value and Internal Rate of Return

Sep 17, 2026

Present value: comparing cash flows that occur at different times, while interest rates tells how to move cash flows in time. Use interest rates to

Ideal Bank: applies the same rate of interest to deposits and loans, and has no transaction costs (real banks have higher rates for loans...). Used for two ways:

- **Compounding:** when cash flow is moved **forward**, **multiplied** by $(1 + r)$ each period.
- **Discounting:** when cash flow is moved **backward**, **divided** by $(1 + r)$ each period.

The Comparison Principle

The ideal bank brings cash flow stream to the same time, then is compared.

- When all cash flows are moved to *present time*, it is called **present value**.
- When all cash flows are moved to the *same future timeline*, it is called **future value**.

General Formula for Present Value

Let $x = (x_0, x_1, x_2, \dots, x_n)$

$$\begin{aligned} PV(x) &= x_0 + \frac{x_1}{(1+r)} + \frac{x_2}{(1+r)^2} + \dots + \frac{x_n}{(1+r)^n} \\ &= \sum_{k=0}^n \frac{x_k}{(1+r)^k} \end{aligned}$$

Note that it's usually assumed that x_0 is negative, for the initial investment.

Main Theorem on Present Value

Two cash flow streams are equivalent for an ideal bank iff they have the same present value.

Internal Rate of Return

Captures the idea on the "return" on an investment.

The return on your investment is the return which would make the present value equal to 0.

Ex. Invest \$100 today, receive \$110 in one year. What is your return?

Intuitively: 10%, why?

$$\begin{aligned} &x(-100, 110) \\ PV(x) &= -100 + \frac{110}{1 + 0.1} = 0 \end{aligned}$$

Def. Let $x = (x_0, x_1, \dots, x_n)$ be a cash flow stream. Then the internal rate of return of this stream

is the number r satisfying:

$$PV(x) = 0$$

$$x_0 + \frac{x_1}{1+r} + \frac{x_2}{(1+r)^2} + \cdots + \frac{x_n}{(1+r)^n} = 0$$

To solve for this r , its more convenient to set

$$c = \frac{1}{(1+r)}$$

and solve

$$x_0 + x_1c + x_2c^2 + \cdots + x_nc^n = 0$$

with the fundamental theory of algebra.

Ex. What is the internal rate of return of $(-2, 1, 1, 1)$?

Solve: $0 = -2 + c + c^2 + c^3$

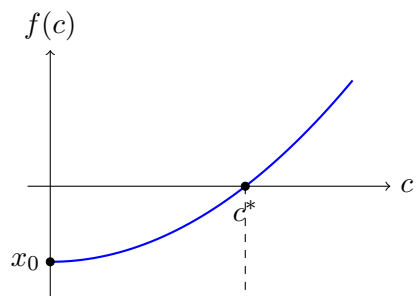
$$c = 0.81 \Rightarrow c = 0.81$$

Main Theorem on IRR

Suppose the cash flow stream $x = (x_0, x_1, \dots, x_n)$ has $x_0 < 0$ and $x_k \geq 0$ where at least one term is strictly positive. Then, there is a unique positive root to the equation

$$x + 0 + x_1c + x_2c^2 + \cdots + x_nc^n = 0$$

Proof: plot $f(c) = x_0 + x_1c + x_2c^2 + \cdots + x_nc^n$



Since $f(0) = x_0 < 0$ and $f'(c) = x_1 + 2x_2c + \cdots + nx_nc^{n-1} \geq 0$ for $c \geq 0$ (as each $x_k \geq 0$, with at least one strictly positive), f is nondecreasing and unbounded above. So f crosses zero exactly once, at some $c^* > 0$.

Net Present Value (NPV)

Computes all present values of *all* cash flows associated with an investment. Includes the investment term x_0 as well.

Ex. Given two cash flow vectors

(1) After 1 year: $(-1, 2)$

(2) After 2 years: $(-1, 0, 3)$

Using NPV solution (choose largest NPV):

$$-1 + \frac{2}{1.1} = 0.82 \quad \text{vs} \quad -1 + \frac{0}{1.1} + \frac{3}{1.1^2} = 1.48$$

NPV says choose option (2).

Using IRR (choose greatest IRR):

$$-1 + 2c = 0 \Rightarrow c = 0.5 \Rightarrow r = 1.0 \quad \text{vs} \quad -1 + 3c^2 = 0 \Rightarrow c = 0.577 \Rightarrow r = 0.7$$

IRR says choose option (1).

IRR and NPV give conflicting recommendations.

When scaled, IRR is scale-invariance, PV is not.

PV also distinguishes between single and repeated cash flows, IRR does not.

In general, NPV is preferred.

Steps to Solving Cash Flow Problems

Lecture 3: Fixed Income Securities

Sep 21, 2026

Def. Financial Instruments

A legal obligation or claim having monetary value (eg. mortgage, stocks and bonds...)

Def. Security

A tradable financial instrument satisfying legal and regulatory requirements.

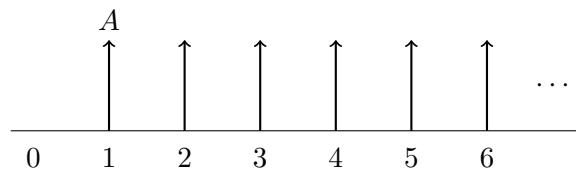
Def. Fixed Income Securities

Securities that promise a fixed income to the holder over some span of time. (eg. bonds, mortgages, annuities)

Annuity

A contract that pays the holder money periodically according to the fixed schedule.

Perpetual Annuity (Perpetuity) - Pays a fixed sum periodically, forever.



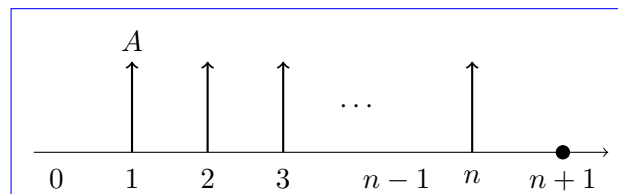
At a per period interest rate of r , the PV is

$$P = \sum_{k=1}^{\infty} \frac{A}{(1+r)^k} = \frac{A}{r}$$

Note: Infinite sum!!

Finite-Life Streams

A stream that pays A for n periods, starting at period 1. What is the PV?

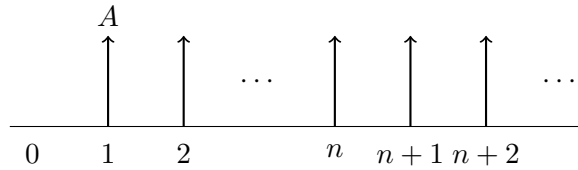


$$P = \sum_{k=1}^n \frac{A}{(1+r)^k} = \frac{A}{r} \left(1 - \frac{1}{(1+r)^n} \right)$$

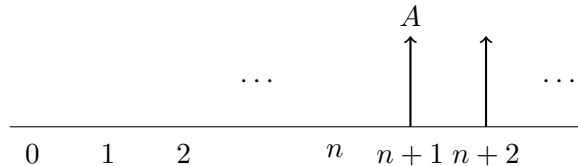
where r is the interest rate per period.

A finite life-stream can be thought as the difference between two perpetuities.

Perpetuity starting at period 1:



Perpetuity starting at period $n + 1$:



$$PV = \frac{A}{r} - \frac{A}{r} \left(\frac{1}{(1+r)^n} \right)$$

$$PV_{fin} = \frac{A}{r} \left(1 - \frac{1}{(1+r)^n} \right)$$

Ex. Loan Calculation

Suppose \$1,000 is borrowed at 12% interest compounded monthly, repayed with equal monthly payments over 5 years. How much are the monthly payments?

Given: $P = \$1,000$ $r = \frac{12\%}{12} = 1\%/mo$ $n = 5 \times 12 = 60$

$$P = \frac{A}{r} \left(1 - \frac{1}{(1+r)^n} \right) \Rightarrow A = \frac{rP}{\left(1 - \frac{1}{(1+r)^n} \right)} = \$22.24/mo$$

Bonds

An agreement to pay money according to the rules of the issue.

Generally, bonds have

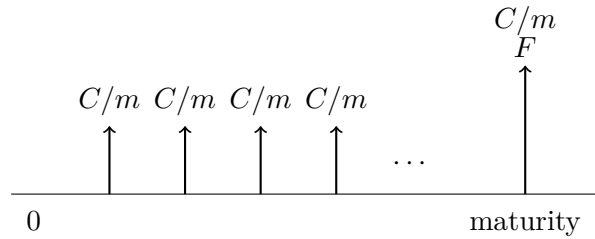
- Face Value, Par Value, Principal - an amount to be paid at the maturity date.
- Coupon Payments - an amount paid periodically, expressed as a % of the face value. Last coupon is paid at maturity.

Ex. Given:

Face Value = \$1,000 Coupon = 9%

The yearly coupon amount is $\$1,000(0.09) = \90

Payoff Structure of a Bond



where C is the annual coupon amount, and F is the face value.

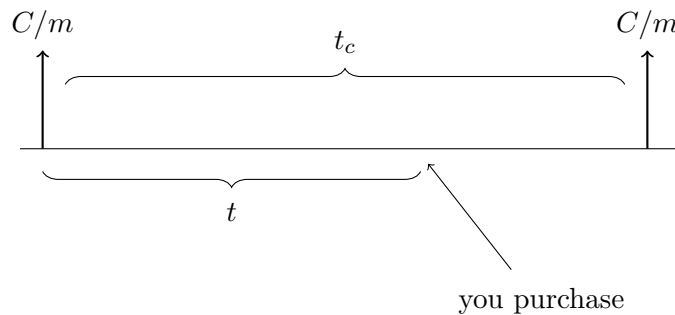
When there are no coupons, the only payoff is the face value at maturity.



This is known as a **zero-coupon bond**, or a zero.

Accrued Interest

Ignored by bond price quotes, but must be added to the price.



You must pay the previous owner their “portion” of the next coupon payment.

$$AI = \frac{\text{days since last coupon}}{\text{days in current coupon period}} \times (C/m) = \frac{t}{t_c}(C/m)$$